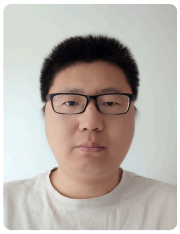
Heng Zhang

Target Role: Embedded Software Engineer, Senior C/C++ Software Engineer

Experience : 16 years
Date of Birth : 1987.12.14
Location : Qingdao, China
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Education

Dalian University of Technology | Software Engineering (Embedded Systems) | B.S. | 2006.09 – 2010.06 | CET-6 (557)

Summary

- 16 years C/C++: 5 years ARM/VxWorks/FreeRTOS embedded, 11 years Linux server networking
- UART, I2C, SPI, USB, TCP/IP; led AirPrint and IPP over USB through Apple certification
- Linux multithreading; FreeRTOS exposure; system design, concurrency, and performance tuning
- Production debugging from application layer through Linux network stack and NIC drivers
- Cross-team technical lead; fluent professional English
- LLM inference and AI agent side projects

Technical Skills

Languages	C/C++ (C++11 at work, C++20 in personal projects); Python, Shell; familiar with Go
Embedded	5 years ARM/VxWorks/FreeRTOS; task scheduling/sync; WindView for RTOS profiling; UART/I2C/SPI/TCP/IP; USB 2.0 device (printer class, descriptors, bulk endpoints), IPP over USB; ARM Cortex-M basics
Networking	TCP/IP and SCTP retransmission/congestion; RFC-based HTTP/UPnP/SNMP on embedded; IPP, Bonjour, Diameter, 3GPP core
Systems	Linux threads, non-blocking I/O, high-concurrency servers; kernel stack and multi-queue NICs; perf, Valgrind, Wireshark
Cloud	Docker, Kubernetes, OpenShift, Helm; C++ service containerization and hitless upgrades
AI	ReAct agents; from-scratch C++ Transformer inference (SIMD, KV cache)

Work Experience

Nokia (China) Qingdao · Senior Software Engineer — 2015.06 – Present

Modules	Owner of three core modules—4G/5G networking, DB cache, middleware; design and quality gate
Leadership	Led 5–6 engineer squads; onboarding and tech sharing
Cloud Native	Core user-database C++ services to Docker/Kubernetes; hitless upgrade (2020 – Present)
Production	NA/EU operator support; T-Mobile incident; Linux stack and virtio multi-queue SCTP issues (2017 – 2020)
Core Network	4G subscriber DB and IMS in C/C++ per 3GPP (2015 – 2017)

Hitachi Information (Dalian) Co., Ltd. · Embedded Software Engineer — 2009.09 – 2015.05

Protocol Stacks	HTTP, UPnP, SNMP, CIFS on ARM/VxWorks/FreeRTOS per RFCs
Print Protocols	AirPrint and IPP over USB; AirPrint 1.1–1.5 certification; Google Cloud Print and scan protocols
Performance	WindView task/sync analysis; Wireshark TCP window and retransmit tuning for scan/transfer throughput
IPv6	IPv6 Ready Logo lab; stack passed Phase-1/Phase-2
Internship	Embedded printer management web UI and automation tests (2009 – 2010)

Project Experience

Apple AirPrint Development & Certification (ARM/VxWorks/FreeRTOS) · Protocol Engineer — 2013.11 – 2015.05

Background Driverless wireless print from phones/tablets/PCs with Apple certification on constrained ARM/VxWorks/FreeRTOS

Contributions
- Discovery: Bonjour (mDNS/DNS-SD) on LAN
- Jobs: IPP submit/status/errors; multi-format print pipeline
- RTOS: C/C++ on VxWorks/FreeRTOS with stack and drivers
- Certification: harness for 1.1–1.5 functional, interop, regression

Results Multiple products certified 1.1–1.5; wireless print shipped

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IPP over USB for Printers (ARM/VxWorks/FreeRTOS) · Lead Developer — 2013.11 – 2015.05

Background USB printing needed host drivers; IPP over USB runs network print protocols on USB printer class for driverless print

Contributions
- USB device: printer class interface and bulk endpoints; HTTP over bulk with framing
- Integration: USB path into existing embedded HTTP server and IPP stack shared with network path
- RTOS: C on VxWorks/FreeRTOS; concurrency between USB I/O and protocol tasks
- Test: host interoperability and compatibility fixes

Results Driverless IPP print and job status over USB matching network behavior

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T-Mobile Core Signaling Latency (Linux Network Stack) · Root-Cause Lead — 2019.06 – 2019.09

Background Rerouted traffic; signaling RTT >140 ms; ~2.29M LTE users; voice call volume −25%

Contributions
- App: packet compare on SCTP; send buffer pile-up; one slow connection blocked peers on same thread
- Kernel: lab replay; NIC queue count, buffers, PM/VM matrix; virtio multi-queue not flow-steering → SCTP reorder/fast retransmit; CWND recovery at high RTT
- Fix: C++ per-connection isolation; buffer/timeout tuning; single-queue, OS upgrade, spreading

Results Restore in 12 h; patch in 72 h; ~10× throughput per connection on single-queue

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User-DB C++ Services on Kubernetes · Lead Developer — 2020.06 – Present

Background User-database C++ services on VMs → Kubernetes with hitless upgrades

Contributions
- Refactor: three core C++ services—single-container multi-process, config inject, hot reload
- Graceful exit: signal handling, link drain, probes; mixed-version rollout

Results 99.9% success during upgrades; 2,500 QPS per instance; T-Mobile, Verizon, Vodafone

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AI Ops Terminal · Solo Developer — 2026.06 – 2026.09

Background Ops needs SSH context; generic AI is blind to terminals

Contributions
- Electron multi-tab SSH with terminal-context AI sidebar
- Function-calling ReAct agent, 30+ tools, three-tier approval

Results Nokia internal AI contest 2nd place; internal rollout

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GPT-2 CPU Inference Engine (C++) · Personal Project — 2026.04 – Present

Background Learn LLM inference via from-scratch C++ Transformer on laptop CPU

Contributions
- AVX2 ops; compute graph; GPT-2 load, KV cache, prefill/decode

Results ~111 tokens/s on 14-thread laptop CPU

Patents, Honors & Certifications

Patent First inventor, PCT WO2025175539A1 (AKMA Authentication Based on Device Information)

Award Oct 2021 Nokia equity grant (1,000 shares)

China Software Designer (Intermediate)

Red Hat OpenShift Application Developer (240-240-408)